Basis for International Trade (Trade Theories)

International Economics - T6

Prof. Kalyan Kolukuluri

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IIM Visakhapatnam

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1. Basis for Trade

Learning Objectives

Objectives

- Understand the *basis for trade* between nations. *Absolute advantage* and *Comparative advantage*.
- Understand the gains from trade and patterns of trade

Basis for Trade

Absolute Advantage

	U.S.	U.K.
Wheat (bushels/hour)	6	1
Cloth (yards/hour)	4	5

- U.S. has an absolute advantage over U.K. in wheat.
- U.K. has an absolute advantage over U.S. in cloth.
- Both nations can gain from specializing in production and then trading.

Goods exchange rate and gains (Aboslute advantage

If the US exchanges **6W** for **6C**. What is:

- USA's gain?
- UK's gain?

Comparative Advantage

	U.S.	U.K.
Wheat (bushels/hour)	6	1
Cloth (yards/hour)	4	2

- U.K. has an absolute disadvantage in both goods.
- Since U.K. labor is half as productive in cloth but six times less productive in wheat compared to U.S., the U.K. has a comparative advantage in cloth.
- U.S. has a comparative advantage in wheat

Comparative Advantage

Comparative advantage law

Even if one nation is less efficient than (has absolute disadvantage with respect to) the other nation in production of both commodities, there is still a basis for mutually beneficial trade.

-David Ricardo

Comparative advantage

How to explain comparative advantage?

Labor theory of value: The value or price of a commodity depends exclusively on the amount of labor used to produce it.

Opportunity cost theory: The opportunity cost of a good is the amount of a second good that must be given up to release just enough resources to produce one additional unit of the first good.

Gains from Trade

1. Both countries can gain from trade.
2. The US gains as long as 6W trade for more than 4 C.
3. The UK gains as long as 2C trades for more than 2 W.
4. Thus the range for mutually beneficial trade is: $4C < 6W < 12C$

Comparative Advantage with Money

- Wages in the two countries will adjust to reflect productivities
- Suppose that the wage rate in the U.S. is \$6/hour, and wages in the U.K. are £1/hour
- In the U.S., 1 worker hour produces 6 bushels of wheat, each bushel of wheat will cost \$1.
- Likewise, in the U.K., 1 worker hour will produce 2 yards of cloth, so a yard of cloth will cost £0.50.
- If the exchange rate is \$1 = £2, we can show each country's prices in U.S. dollars.

Comparative Advantage in Monetary terms

	U.S.	U.K.
Unit Price Wheat	\$1	\$2
Unit Price Cloth	\$1.5	\$1

- The table reflects an exchange rate of $\$1 = £2$
- For mutually beneficial trade, the price of a bushel of wheat must be between $\$1 < P_W < \$2$
- Changes in exchange rates will change gains from trade and possibly patterns of trade.

PPC- Production Possibilities Curve with Constant costs

PPC

Production Possibilities Curve/Frontier shows alternative combinations of the two commodities a nation can produce by fully using all resources with the best available technology.

Without trade, the PPC also represents its *consumption frontier*.
Constant opportunity costs arise when:

1. Resources are either perfect substitutes for each other or used in fixed proportion in the production of both commodities,
2. All units of the same factor are homogeneous.

Production Possibilities (Pre and Post-trade)

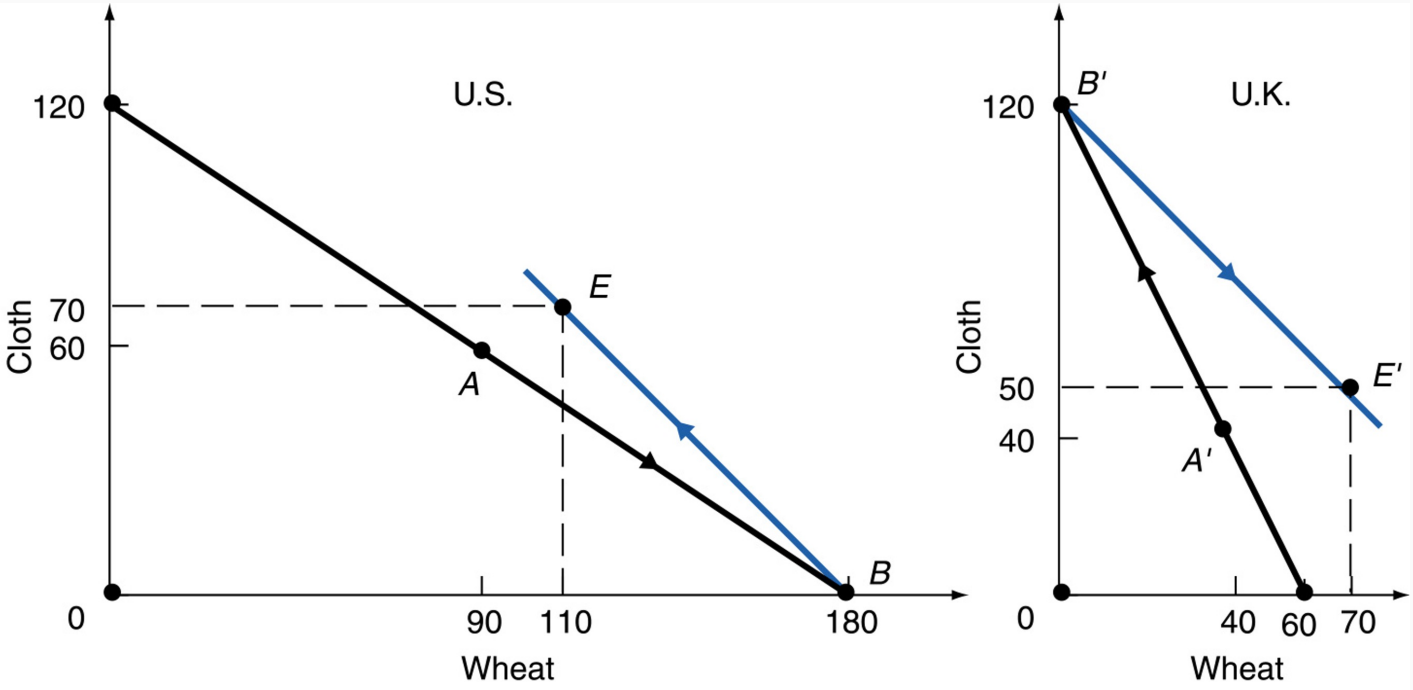

Figure 1: Pre and Post-trade Production Possibilities

Demand and Supply - Post Trade

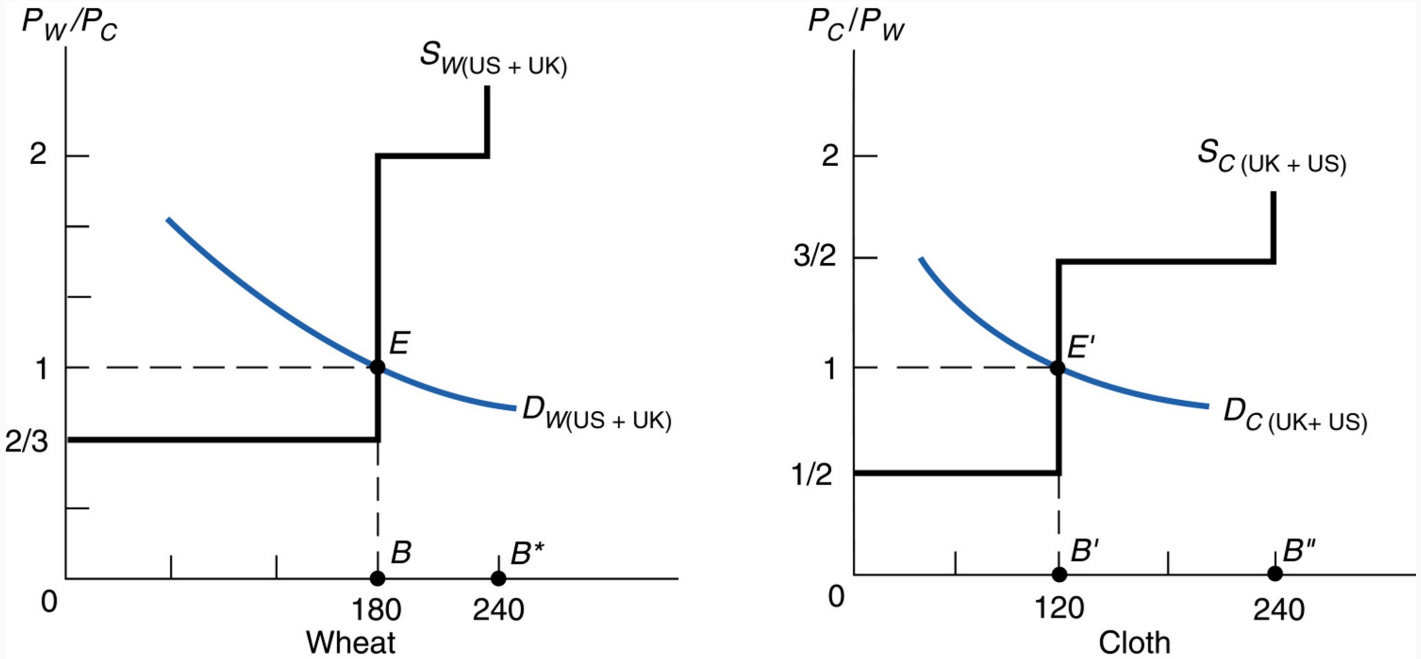

Figure 2: Demand and Supply in a Free Trade set up

Conclusion

Under constant cost conditions, nations will completely *specialize* in their comparative advantage .

With complete specialization in both nations, the equilibrium-relative commodity price of each commodity lies between the pretrade relative commodity price in each nation.

The End